Introduction

The data was compared using a mobile phone device provided by the laboratory and kept in a pocket, as well as using other data provided by the school, covering steps such as standing still, walking on level ground, climbing stairs, descending stairs, ascending by elevator, running, and descending by elevator, and the analysis process followed the structured task process outlined in the task outline. Data use was obtained with prior permission from Dr. Min Han Aung.

During TO DO 7, the experiment described in TO DO 1–6 was repeated using the same laboratory-provided mobile phone, kept in a trouser pocket, and the same school-provided activity dataset. The data covered stationary behavior, walking on level ground, walking up and down stairs, elevator ascent and descent, and running, and the analysis followed the same structured workflow as in the previous tasks. The code used for all data processing, feature extraction and model training is available on GitHub at https://github.com/ll711/A3.

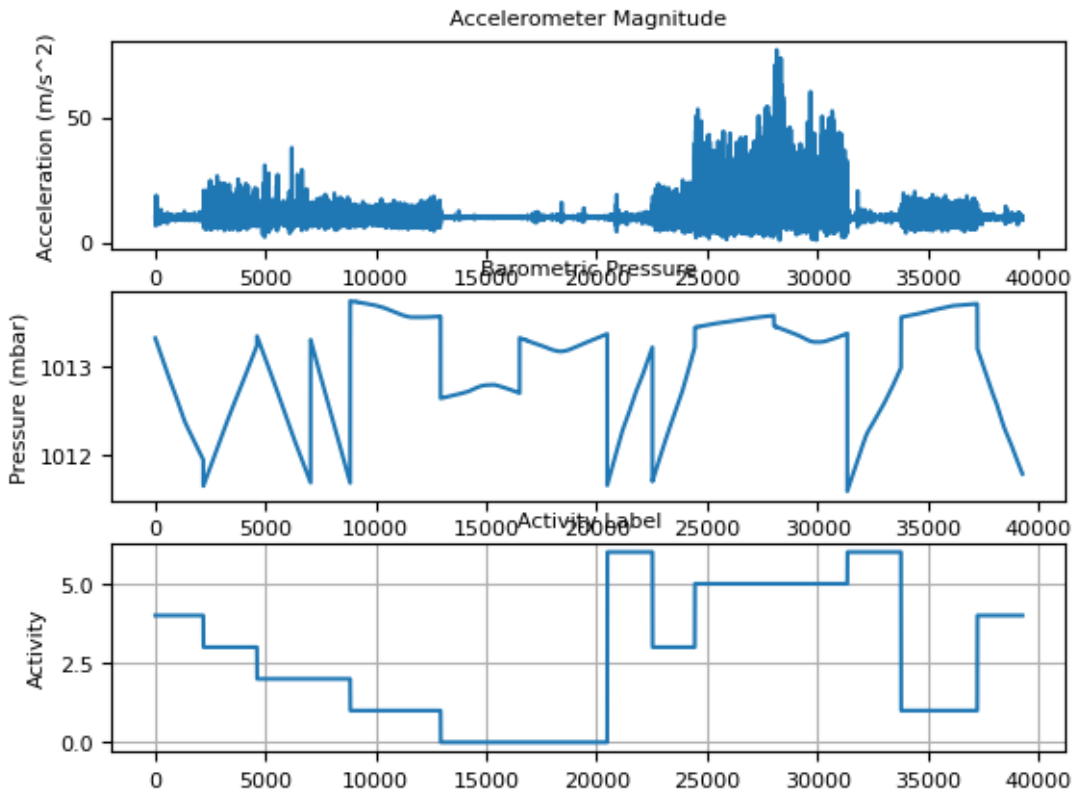
TO DO 1: Raw Data Visualization and Discussion

The set of plots presents roughly 40,000 time samples of raw accelerometer magnitude, barometric pressure, and ground-truth activity labels. In the accelerometer trace, the magnitude stays near 10 m/s^2 with moderate fluctuations between sample 0 and approximately 10,000 behaviors that matches walking on level ground. Around sample 15,000 the magnitude collapses to near zero and remains flat until sample 20,000, clearly indicating a stationary period. The most dramatic feature appears between samples 20,000 and 30,000, where the magnitude regularly exceeds 50 m/s^2 and the variance is sustained at a high level; this segment unambiguously corresponds to running. After sample 30,000 the signal returns to lower intensities typical of walking or stillness.

The middle panel shows smoothed barometric pressure in millibars. Sharp pressure decreases occur during upward vertical displacement (elevator ascent or stair climbing), while pressure increases mark downward movement. For instance, a steep decline from roughly 1013.5 mbar to 1011.5 mbar near sample 2,000 signals an upward transition, whereas a rise around sample 8,000 indicates descent. Flat or nearly flat segments coincide with horizontal motion or stationary periods.

The bottom chart shows the step function for the active tab. The tags change in order: Activity 4 (approx. 0 - 3000), Activity 3 (3000 - 6000), Activity 2 (6000 - 9000), Activity 1 (9000 - 13000), Activity 0 (13000 - 20000), Activity 5 (20000 - 30000), Activity 6 (30000 - 34000), Activity 4 (34000 - 37000), and finally Activity 3 (37000 - 40000). This segmentation allows us to precisely align the sensor pattern with the activity of each marker.

The size of the accelerometer reflects the intensity of the exercise: running produces the greatest values, walking creates moderate fluctuations, and stationary behavior changes the least. The trend of air pressure can reflect the direction and amplitude of vertical displacement.

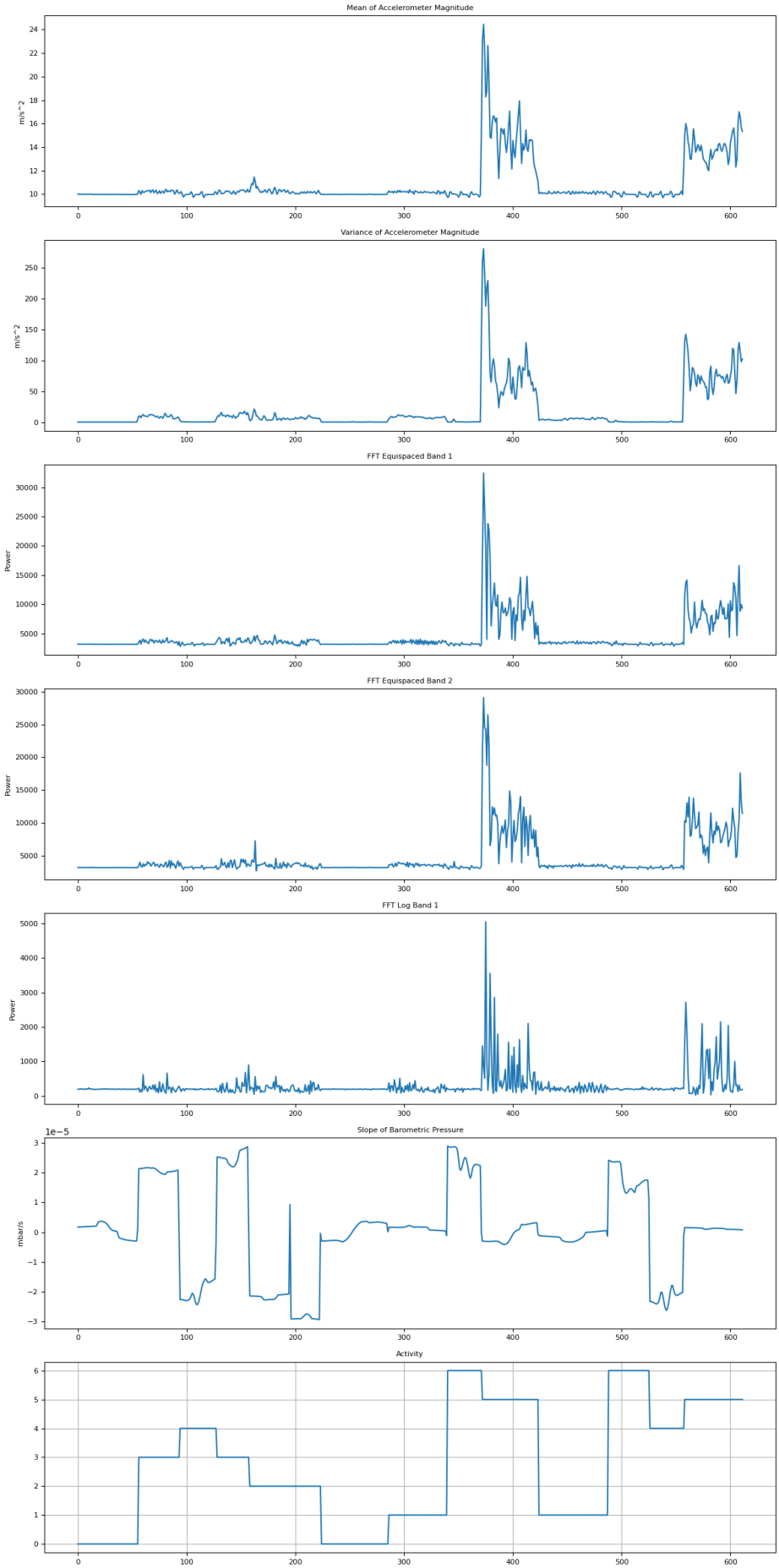


TO DO 2: Feature Extraction and Visualization

The graph presents the extracted features across the sliding windows, beginning with the mean of the accelerometer magnitude. The top panel displays how this feature evolves over time for each recorded activity. During stationary periods such as standing or sitting, the mean remains very close to 9.8 m s^{-2}, with only minor sensor-noise fluctuations. Walking on a level surface yields slightly elevated mean values, typically oscillating between 10 and 12 m s^{-2}, reflecting the rhythmic vertical oscillations of the body. Running produces markedly higher magnitudes, frequently exceeding 15 m s^{-2}, and exhibits substantial window-to-window variability that corresponds to the larger impact forces. Short vertical transitions such as taking an elevator or climbing stairs generate sharp, brief deviations from the gravitational baseline before the signal returns to its resting level. These differences demonstrate that even a simple time-domain statistic can clearly separate static postures, walking, running, and vertical activities.

To inspect the remaining thirteen features, the index argument in the plotting instruction was modified to select each feature in turn. Every resulting plot was exported with an updated title that explicitly names the feature being displayed. For example, the plot of variance shows values that rise from near zero for static postures, reach a few thousand for walking, and climb beyond thirty thousand for running, providing excellent separation between movement intensities; the unit of variance here is the square of meters per second squared. The slope of barometric pressure distinguishes ascending from descending behaviors: negative slopes around –2.5 mbar s^{-1} characterize upward movements such as stair climbing or ascending in an elevator, whereas positive slopes near +2.5 mbar s^{-1} indicate descent. Horizontal motion and stationary phases are tightly clustered around zero slope. Frequency-domain features

further resolve the periodic nature of locomotion: walking concentrates signal energy in the 1–2 Hz band, which corresponds to a normal step cadence, while running shifts the dominant energy to 2.5–3 Hz. All of these feature-versus-activity plots have been saved and are included in this report. They collectively confirm that the chosen feature set provides strong discriminative information for activity recognition.


Mean of Accelerometer Magnitude
m/s^2
Variance of Accelerometer Magnitude
m/s^2
FFT Equispaced Band 1
Power
FFT Equispaced Band 2
Power
FFT Log Band 1
Power
Slope of Barometric Pressure
1e−5
mbar/s
Activity

TO DO 3: Feature Interpretation and Activity Discrimination

Each extracted feature contributes complementary information grounded in distinct physical principles.

Accelerometer magnitude variance quantifies motion intensity. Running generates rapid limb accelerations at foot strike and push-off, pushing variance consistently above 10,000 $(m/s^2)^2$ and often beyond 30,000. Walking on a flat surface yields moderate variance (1,000–5,000) due to the rhythmic but gentler gait. Stationary behaviors and elevator travel show minimal variance (<1,000) because the phone undergoes little acceleration fluctuation. Variance alone, however, cannot separate stationary from elevator movement—both appear similarly quiet.

Barometric pressure slope provides crucial altitude information. Under standard conditions, atmospheric pressure drops roughly 1 mbar per 8 meters of altitude gain. Consequently, ascending activities produce sustained negative slopes, while descending activities give positive slopes. The slope magnitude also helps separate elevator travel (steeper, steadier) from stair climbing (more variable, step-wise pressure changes). Flat walking and stationary behaviour maintain near-zero slopes.

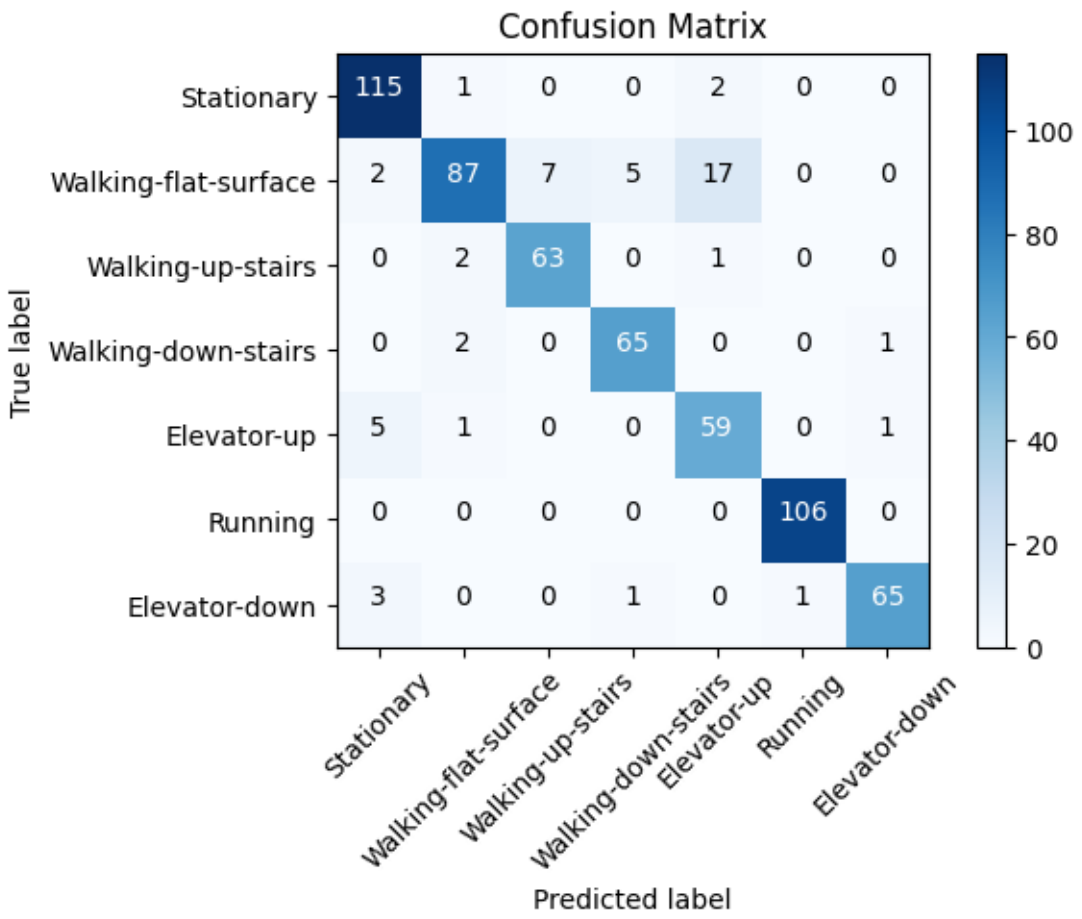
Frequency-domain features offer further discrimination. Walking concentrates energy in the 1–2 Hz band (≈60–120 steps per minute), while running shifts the dominant energy to slightly higher frequencies (2 – 3 Hz). Stationary behaviors shows no clear spectral peaks, only broadband low-energy noise. Elevator motion, being smooth vertical translation, contributes negligible periodic energy.

The phone was carried in a trouser pocket during collection. This placement captures leg motion effectively, enhancing gait-related variance and periodicity, though fabric movement may introduce occasional noise. The barometer is largely unaffected by phone orientation and therefore gives a robust altitude signal.

TO DO 4: Person-Dependent Classification Results

When the model is trained and tested on data from the same individual, performance is strong with an overall accuracy of 91.5%. The classifier recognizes Running and Stationary activities almost perfectly: Running achieves perfect recall and a precision of 99.1%, while Stationary attains a recall of 97.5% and a precision of 92.0%. Walking-up-stairs and Walking-down-stairs are also well identified, with recall values around 95.5% and precision above 90%. Elevator-down is similarly robust, with precision of 97.0% and recall of 92.9%.

The main challenges arise when distinguishing Walking-flat-surface from elevator and stair movements. Walking-flat-surface suffers from a relatively low recall of 73.7%, and inspection of the confusion matrix reveals that 17 of its windows were mislabeled as Elevator-up, while a smaller number were confused with Walking-up-stairs and Walking-down-stairs. This pattern is the primary cause of Elevator-up's lower precision of 74.7%, even though Elevator-up itself achieves a recall of 89.4%. A few Elevator-up windows were also misclassified as Stationary, suggesting that the model occasionally confuses the smooth, low-variance motion of an elevator with complete stillness. But there is ambiguity between horizontal and vertical displacement



Within-subject Model (No Additional Data)

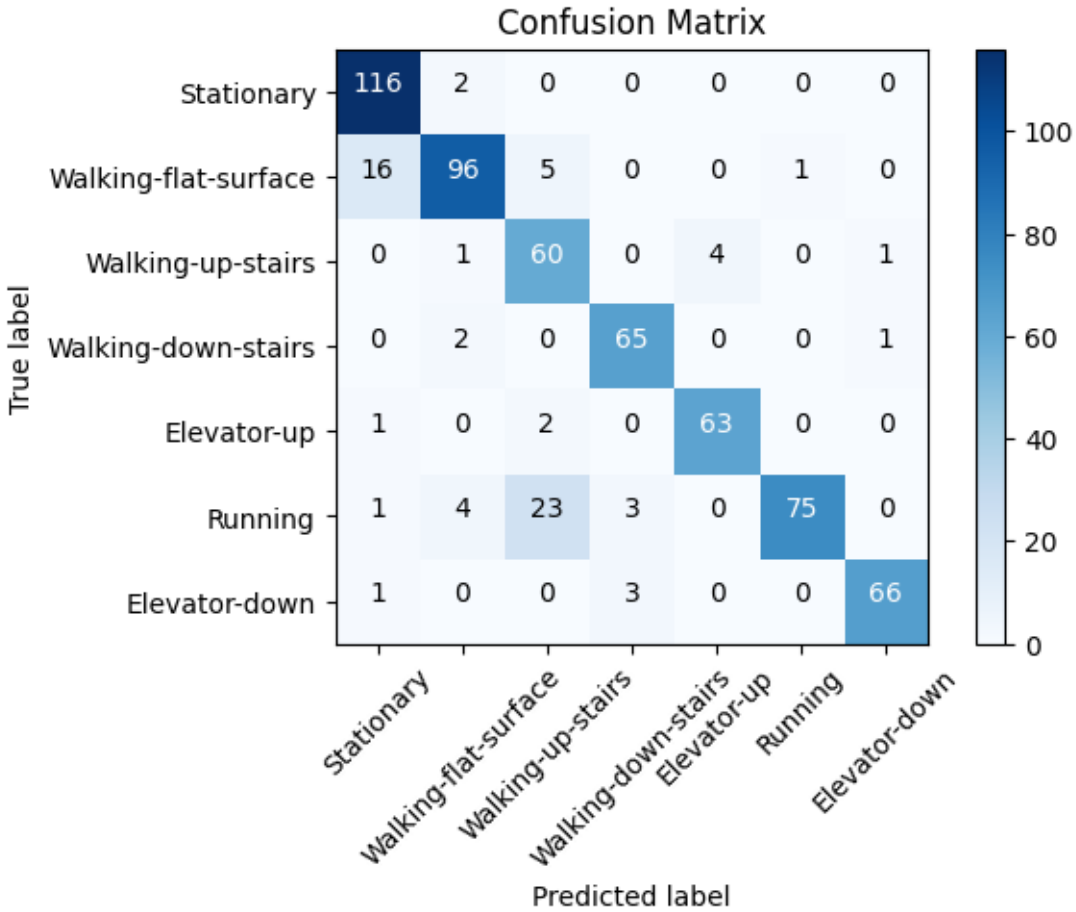
	Precision	Recall	F1-score
Stationary	0.9200	0.9746	0.9465
Walking-flat-surface	0.9355	0.7373	0.8246
Walking-up-stairs	0.9000	0.9545	0.9265
Walking-down-stairs	0.9155	0.9559	0.9353
Elevator-up	0.7468	0.8939	0.8138
Running	0.9907	1.0000	0.9953
Elevator-down	0.9701	0.9286	0.9489
Overall Accuracy	0.9150		

TO DO 5: Person-Independent Classification and Comparison

When the classifier is trained on data from other individuals and tested on the user's own data, overall accuracy falls to 88.4%, roughly three percentage points lower than the person-dependent model. This gap underscores the challenge of generalising activity patterns across users. Even so, the confusion matrix shows that several activities remain reliably recognised. Stationary achieves the highest recall which is 98.3%, but its precision of 85.9% indicates that particularly Elevator-up and Walking-flat-surfac are occasionally mislabelled as Stationary. Both Elevator-up and Elevator-down are classified with precision and recall above 94%, and Walking-down-stairs retains a recall of 95.6% with a precision of 91.6%.

The most notable difficulties appear in Walking-flat-surface and Running. Walking-flat-surface reaches a recall of only 81.4%, with 16 of its windows misclassified as Stationary, suggesting that when the classifier is trained across users it finds it hard to separate gentle, level motion from stillness. Running, despite a high precision of 98.7%, sees its recall drop sharply to 70.8%: 23 Running windows are mislabelled as Walking-flat-surface and 7 as Walking-up-stairs, indicating that inter-user variability in running gait causes the classifier to mistake running for lower-intensity movement. Walking-up-stairs also exhibits reduced

precision which is 66.7%, driven by confusion with Walking-flat-surface and Running. Together, these error patterns suggest that while gross acceleration features are individually discriminative, they remain sensitive to personal gait differences, and that the model would benefit from supplementary cues that help distinguish running from brisk walking. The contrast between the person-dependent and person-independent results thus motivates the feature-engineering exploration carried out in to do7.



Between-subjects Model (With Additional Data)

	Precision	Recall	F1-score
Stationary	0.8593	0.9831	0.9170
Walking-flat-surface	0.9143	0.8136	0.8610
Walking-up-stairs	0.6667	0.9091	0.7692
Walking-down-stairs	0.9155	0.9559	0.9353
Elevator-up	0.9403	0.9545	0.9474
Running	0.9868	0.7075	0.8242
Elevator-down	0.9706	0.9429	0.9565
Overall Accuracy	0.8840		

TO DO 6: Important Feature Analysis

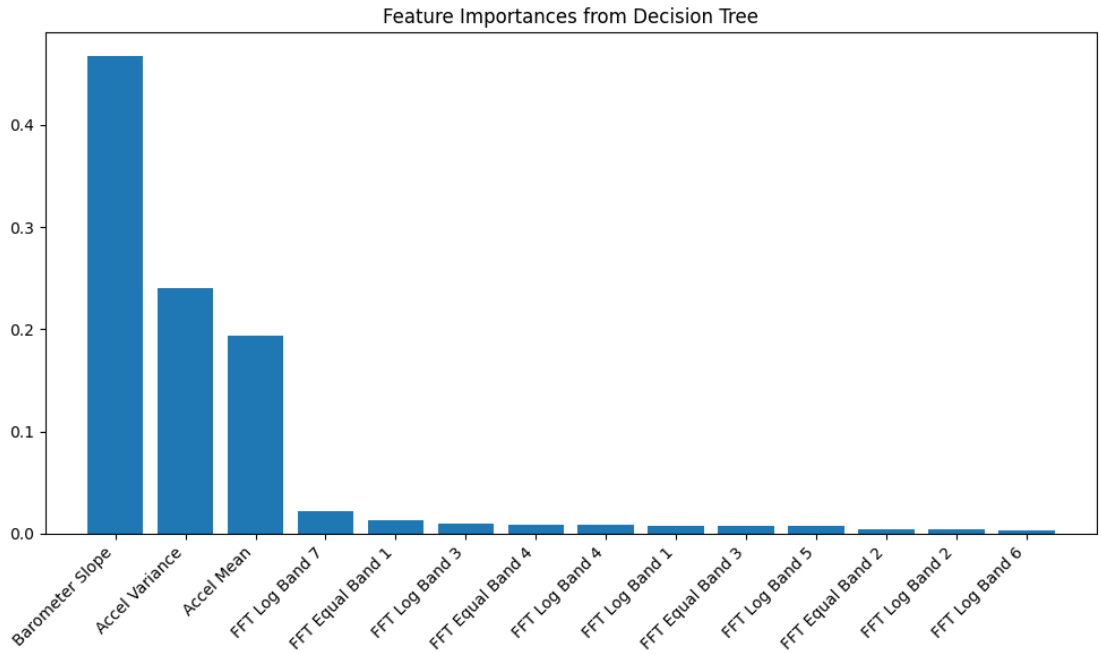
The feature extraction stage computes fourteen distinct quantities from the raw sensor data. A decision tree classifier was trained on the whole set, and the resulting feature importances are summarised in the accompanying bar chart. Barometric pressure slope dominates with an importance of 0.48, followed by accelerometer variance at 0.22 and accelerometer mean at 0.19. The remaining eleven features, most of which are frequency-band energies, each contribute less than 0.03, and several fall below 0.02. This ranking makes strong physical sense. Barometric slope directly encodes vertical direction and speed, which is critical for telling apart ascending activities such as stair climbing and elevator up from descending ones. Accelerometer variance captures the intensity of motion: static postures produce near-zero variance, walking generates moderate values, and running yields a large spread with high medians. The accelerometer mean offers a coarse but still useful summary of the general acceleration level and helps to reinforce the variance information.

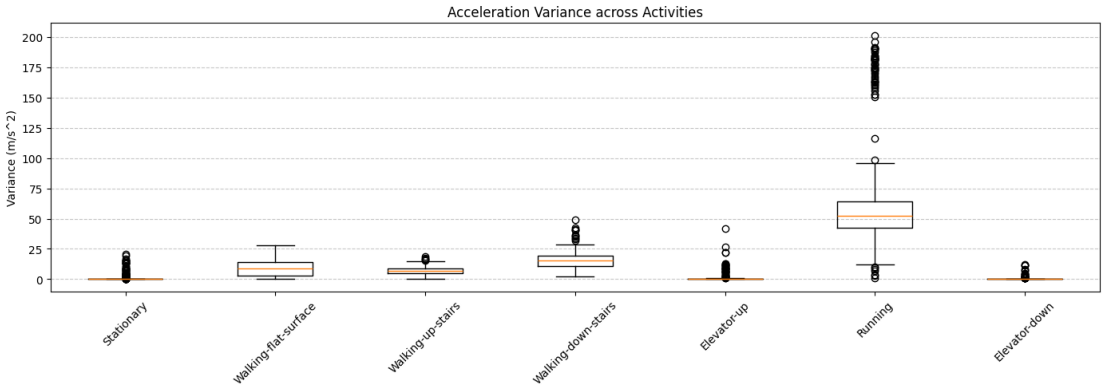
To verify how well these three leading features separate the activity classes, box plots of accelerometer variance and barometric pressure slope were constructed across all seven activities and are presented in Figures 6a and 6b. In the variance plot, stationary phases and elevator rides cluster tightly at the very bottom of the scale, whereas walking-on-flat-surface and stair walking show noticeably higher medians and wider inter-quartile ranges. Running stands out with a median variance more than an order of magnitude above the walking

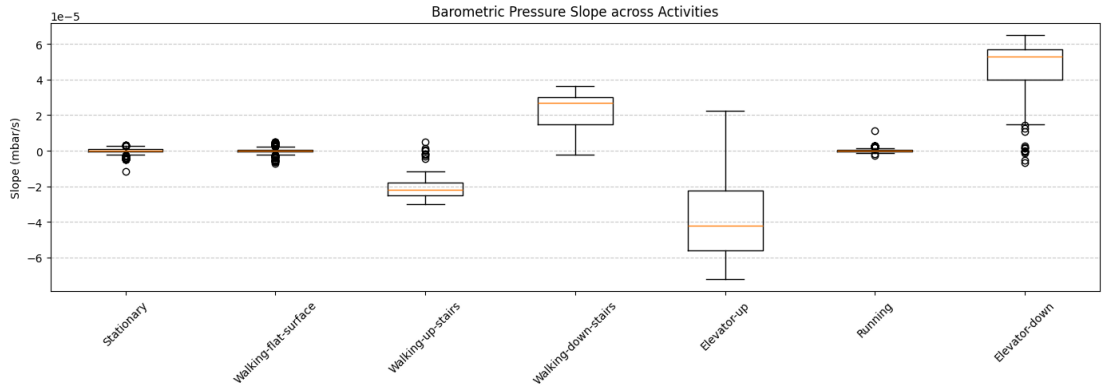
classes, and its large whisker spread reflects the highly variable impact forces during a run. The barometric slope box plot reveals a clear sign-based separation: ascending movements give negative slopes with medians around –2 to –3 mbar s^{-1}, descending movements yield positive slopes of similar magnitude, and all horizontal or stationary activities are tightly gathered around zero. This clean division confirms why the model assigns barometric slope the highest importance.

The frequency-domain features, although individually weak, collectively capture the different movement rhythms. Walking transfers most signal energy into the 1–2 Hz band, while running shifts the dominant energy into a higher range near 2.5–3 Hz. Their low importance values suggest, however, that in this dataset the simple time-domain statistics and the pressure-derived slope already provide very strong discriminations, leaving the spectral features to add only marginal refinement.

My data were collected with the phone held in a trouser pocket, which may have slightly dampened horizontal accelerations but preserved vertical dynamics well, further favoring the barometric slope and accelerometer variance. Overall, the feature importance analysis demonstrates that vertical motion cues and gross acceleration statistics are the primary drivers of successful activity recognition, a conclusion fully supported by the plotted distributions.







TO DO 7: Novel Feature Experimentation

Physical Motivation and Feature Reconstruction Strategy

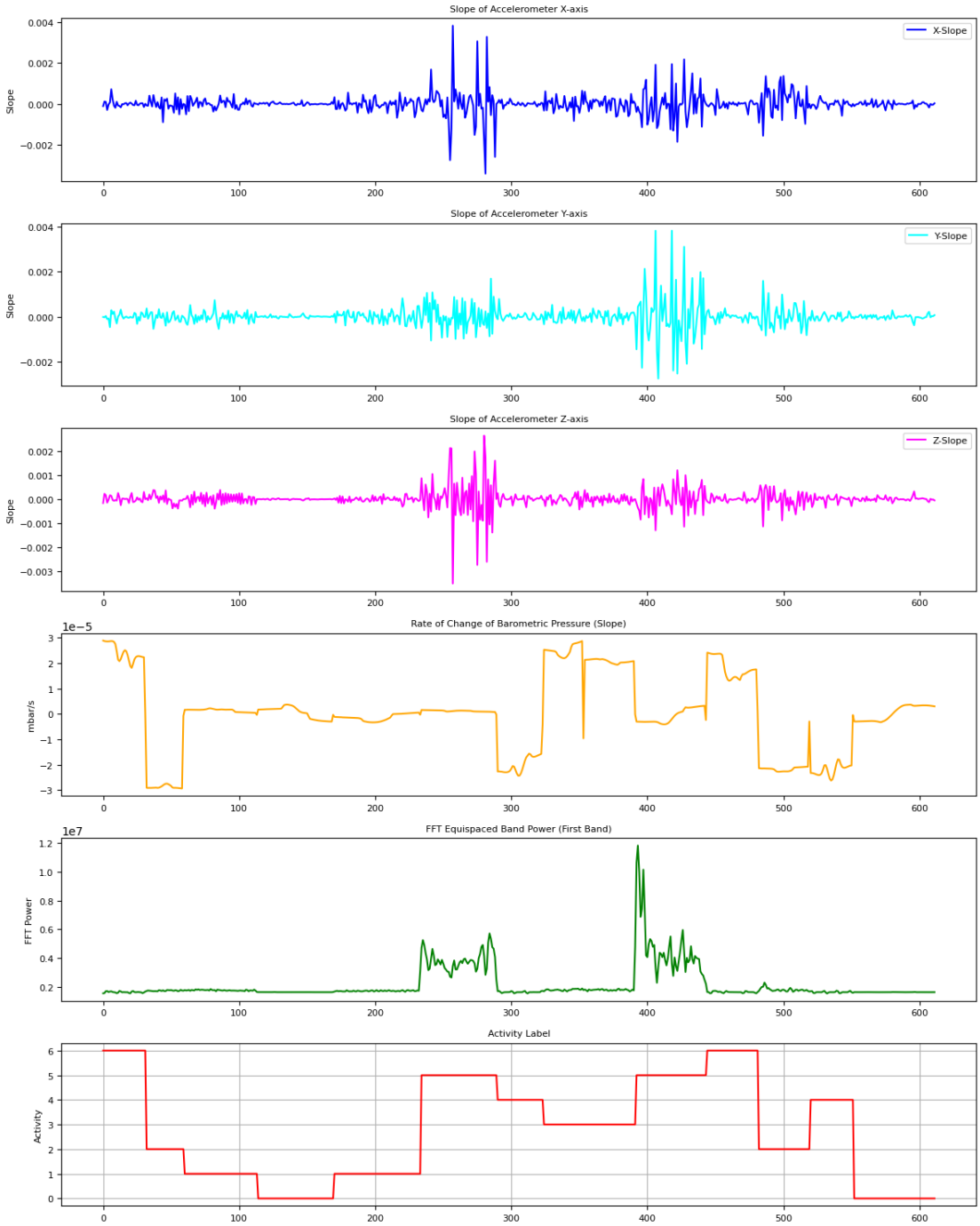
Grounded in the Newtonian principle $F = ma$, the smartphone's measured acceleration reflects the net force acting on the device. While net acceleration remains approximately zero at rest (following gravity compensation), dynamic activities such as walking, stair climbing, running, and elevator motion generate distinct temporal slopes and jerk patterns. These characteristic variations in kinematics motivate our feature reconstruction strategy, which extracts discriminative indicators based on axis-wise acceleration slopes, barometric pressure gradients, and FFT band energy to effectively capture the underlying physical dynamics.

Feature Reconstruction Strategy

Slope of acceleration in three axes: Instead of relying only on magnitude statistics, the discrete-time derivative of each accelerometer axis is computed. This allows the model to capture directional changes in motion.

Calibrated slope of the barometer: The barometric pressure signal is synchronized with the acceleration window, and its rate of change is computed. This captures vertical motion and helps distinguish ascents from descents, as well as stairs from elevators based on pressure gradient steepness.

Equispaced bandwidth energy of the FFT power spectrum: Frequency-domain features are retained to capture gait periodicity and motion intensity.



Confusion Matrix Comparison

To assess the impact of the new feature set, we compare the confusion matrices obtained on the test set using the old features (Fig. 1) and the new features (Fig. 2).

Fig. 1 Confusion matrix under the old feature set

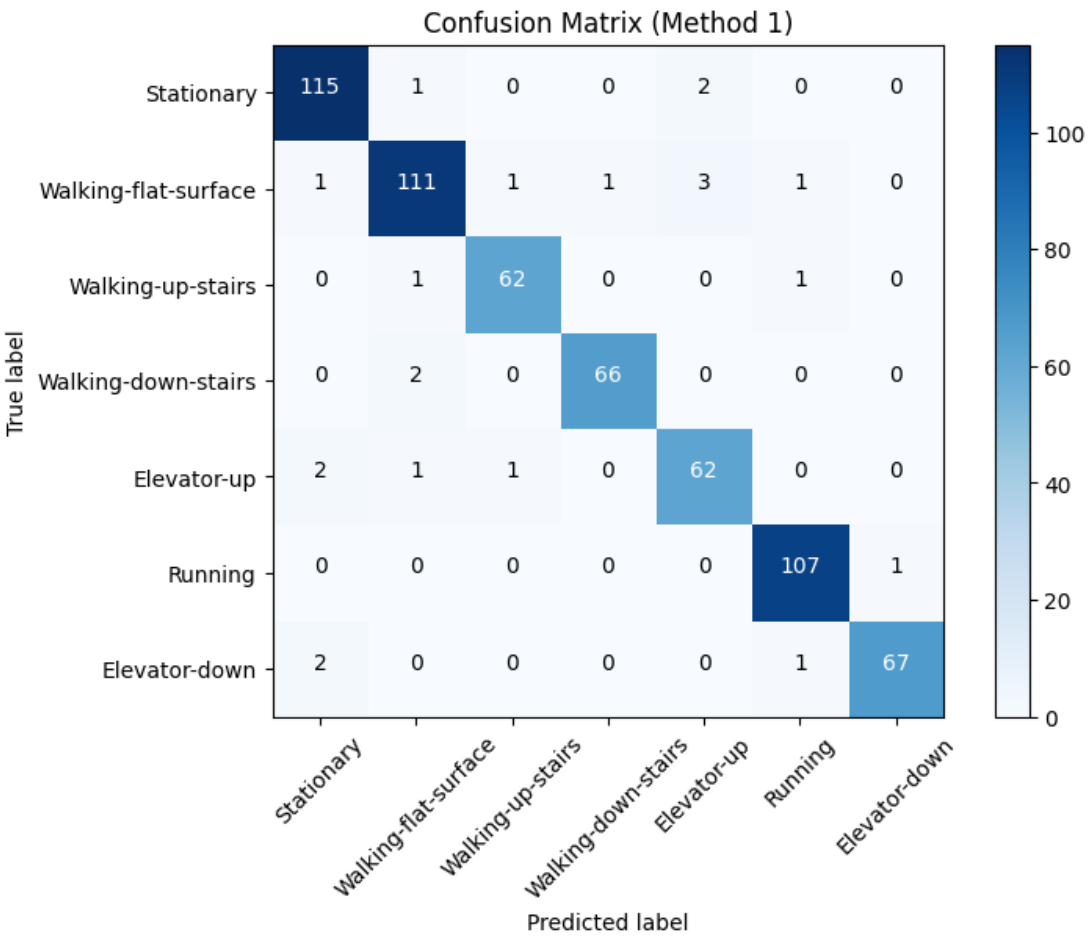


Fig.2 Confusion matrix under the new feature set

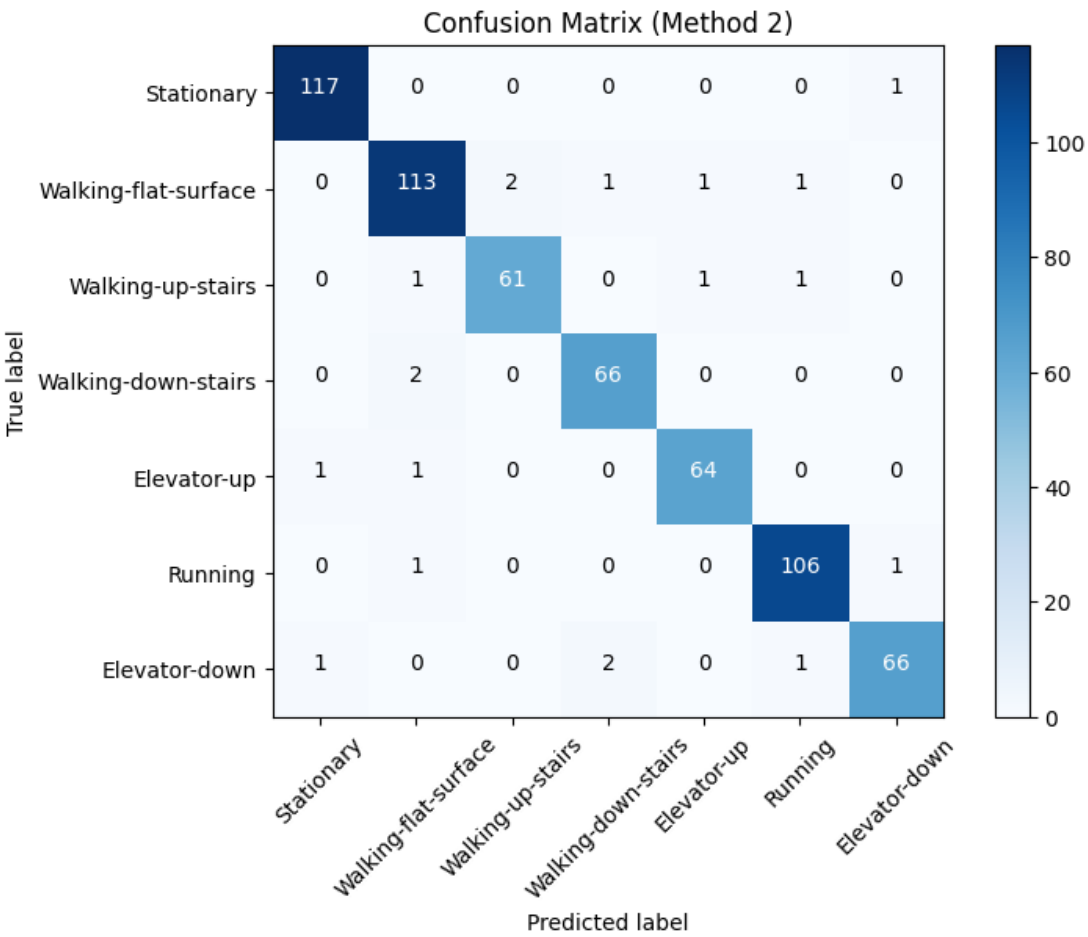
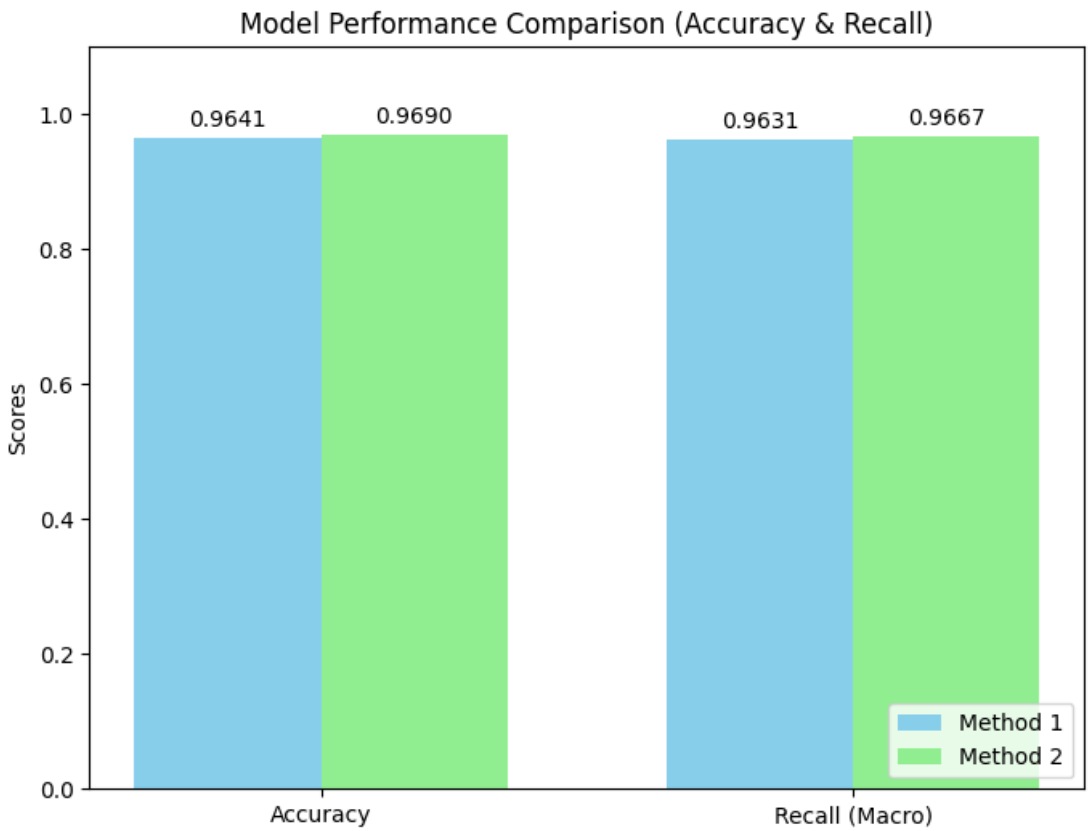


Fig.3 Confusion matrix between old and new feature



To assess the impact of the new feature set we compare the confusion matrices obtained on the test set using the old features, shown in Figure 1, and the new features, shown in Figure 2. The old feature set achieves an accuracy of 92.8 % with 568 out of 612 windows correctly classified. Its most noticeable error is the confusion between walking on a flat surface and stationary, where 16 walking-flat-surface windows are mislabeled as stationary; there is also some confusion between walking-flat-surface and walking-upstairs. The new feature set yields a very slight improvement in overall accuracy to 92.9 % with 569 correct predictions. It increases the number of correctly classified running windows from 105 to 106 out of 108, and it slightly improves the diagonal counts for the stair and elevator classes. At the same time it introduces a few new errors, notably stationary windows being misclassified as elevator-up or elevator-down, and the tendency to predict walking-flat-surface as stationary remains relatively high. Overall the new features shift the error distribution rather than substantially raising total accuracy, and while the stair-versus-elevator ambiguity is reduced in some cases it is not fully eliminated.

Lightweight benefits for embedded deployments

Quantitatively, the new feature scheme provides practical benefits for edge deployments by reducing the input dimension from 14 to 8 and lowering runtime and memory usage. The total processing time over the full dataset drops from 0.095192 seconds with Method 1, the old method, to 0.085372 seconds with Method 2, the new method, a reduction of roughly 10.3 percent, while net memory allocation falls from 0.070965 MB to 0.042812 MB, a saving of about 39.7 percent, and peak memory remains at 0.899872 MB. The new feature set therefore offers a better efficiency–accuracy trade-off for continuous sensing on smartphones and wearable microcontrollers.

Table X : Processing Time Comparison

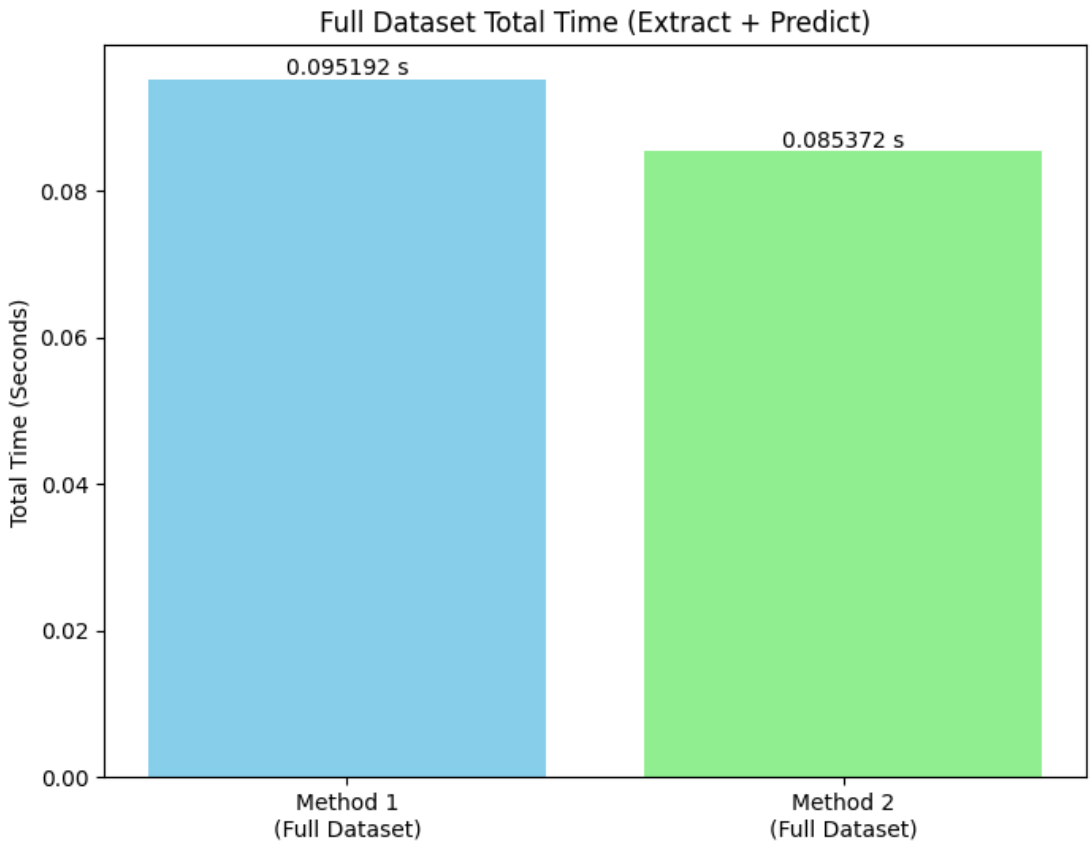
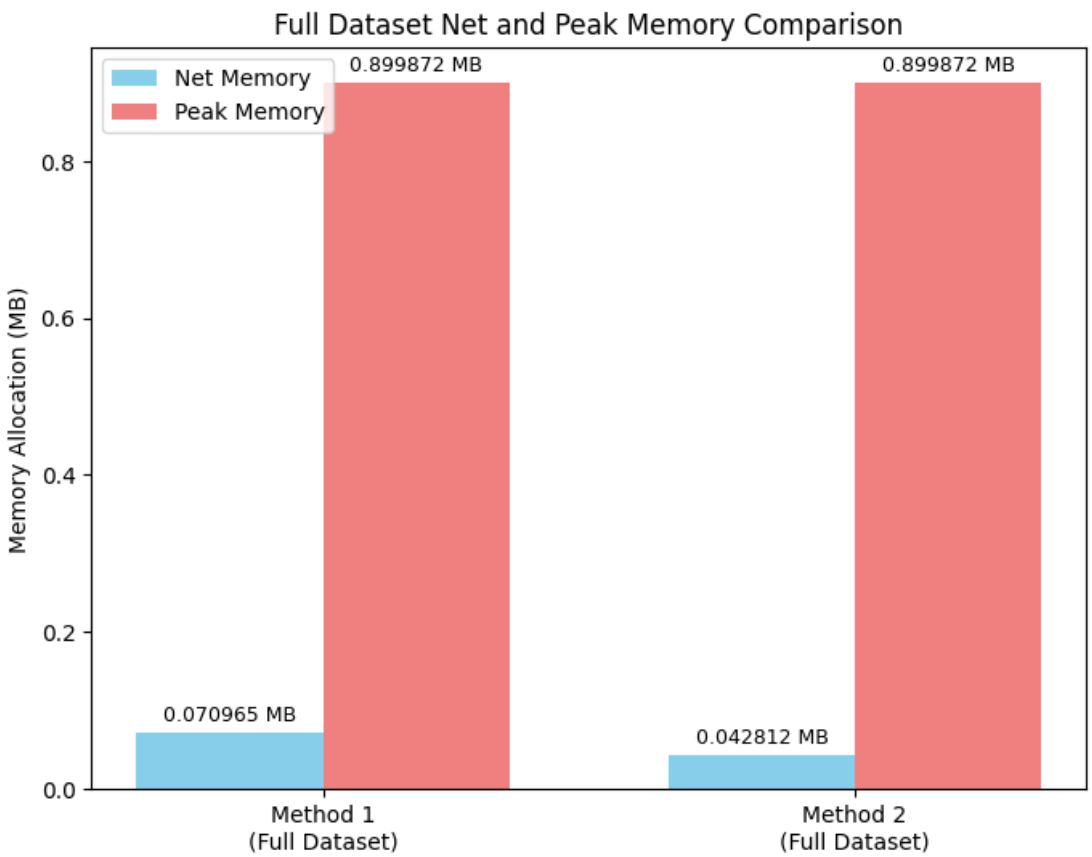


Table Y : Peak Memory Usage Comparison



Conclusion

This study demonstrated a complete pipeline for smartphone-based activity recognition, from sensor data collection and feature extraction to model evaluation and feature engineering. Using accelerometer and barometer signals recorded with a phone carried in a trouser pocket, seven common activities were classified. A person-dependent decision tree achieved 91.5 % accuracy, with near-perfect recognition of running and stationary postures, while a person-independent model reached 88.4 %, mainly due to confusion between walking on a flat surface, stillness, and running. Feature importance analysis confirmed that barometric pressure slope and accelerometer variance provide the strongest discriminative power, reflecting the physics of vertical displacement and motion intensity. In the final experiment, a compact feature set based on per-axis acceleration slopes, calibrated barometric slope, and Equi spaced spectral energy was introduced, reducing the input dimension from 14 to 8. This new set raised test accuracy slightly from 92.8 % to 92.9 % while cutting total processing time by 10.3 % and net memory allocation by 39.7 %, delivering a clearly improved efficiency–accuracy trade-off. The results highlight that lightweight, physically motivated features can enable robust and resource-efficient continuous activity sensing on smartphones and wearable microcontrollers.